Byeong-Il Ham

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Education

Korea Advanced Institute of Science and Technology, Robotics — Daejeon, KR, 2024 – present

Korea Advanced Institute of Science and Technology, Robotics — Daejeon, KR, 2022 – 2024
- Robust Swing Leg Planning Algorithm for Quadruped Robots Using Dual Convex Model-Predictive Control

University of Kwangwoon, School of Robotics — Seoul, KR, 2016 – 2022

Experience

IEEE, Graduate Student Member — Mar 2025 – present, 1 year 8 months

IEEE Robotics and Automation Society, Technical Committee Member — Jan 2026 – present, 10 months
- Model-Based Optimization for Robotics
- Humanoid Robotics
- Robot Control
- Robot Learning

Awards

Best System Paper Award — Dec 2024
Hyun-Bin Kim, Chanseok Kim, Byeong-Il Ham, Jeonguk Kang, Minseong Choi, Keun-Ha Choi, Kyung-Soo Kim

Publications

A Compact Optical Joint Torque Measurement System Using Redundant Photo-Reflectors and Quadratic-Programming Calibration
Hyun-Bin Kim, Byeong-Il Ham, Kyung-Soo Kim

Accelerated Generation of Smooth Human-Aware Robot Trajectories via Consistency Distillation from a Diffusion Model
Byeong-Il Ham, Hyun-Bin Kim, Kyung-Soo Kim

Dual-MPC Footstep Planning for Robust Quadruped Locomotion
Byeong-Il Ham, Hyun-Bin Kim, Jeonguk Kang, Keun Ha Choi, Kyung-Soo Kim
doi.org/10.1109/TMECH.2026.3720700

Ground-Aware Octree-A *Hybrid Path Planning for Memory-Efficient 3D Navigation of Ground Vehicles**
Byeong-Il Ham, Hyun-Bin Kim, Kyung-Soo Kim
ieeexplore.ieee.org/abstract/document/11301310

Adaptive Gait Pattern Switching under External Disturbances Using Multi-Modal MPC
Jeonguk Kang, Byeong-Il Ham, Seungho Han, Hyun-Bin Kim, Kyung-Soo Kim
ieeexplore.ieee.org/document/11217651

Compact and Optimized Optical Six-Axis Force/Torque Sensor for Legged Robots
Hyun-Bin Kim, Byeong-Il Ham, Keun-Ha Choi, Kyung-Soo Kim
ieeexplore.ieee.org/document/11194133

A Miniature High-Resolution Tension Sensor Based on a Photo-Reflector for Robotic Hands and Grippers
Hyun-Bin Kim, Byeong-Il Ham, Kyung-Soo Kim
ieeexplore.ieee.org/abstract/document/11150551

Development of Remote Piping Inspection System with Dual-Mode Locomotion Quadruped Robot
Hyun-Bin Kim, Chanseok Kim, Byeong-Il Ham, Jeonguk Kang, Minseong Choi, Keun-Ha Choi, Kyung-Soo Kim
link.springer.com/chapter/10.1007/978-3-031-92011-0_25

Temperature Compensation Method for a Six-Axis Force/Torque Sensor Using a Gated Recurrent Unit
Hyun-Bin Kim, Seokju Lee, Byeong-Il Ham, Keun-Ha Choi, Kyung-Soo Kim
ieeexplore.ieee.org/document/11021311

External force adaptive control in legged robots through footstep optimization and disturbance feedback
Jeonguk Kang, Hyun-Bin Kim, Byeong-Il Ham, Kyung-Soo Kim
ieeexplore.ieee.org/document/10731682

Reviewing

IEEE Sensors Journal, Reviewer 2026
2026

IEEE Transactions on Industrial Electronics (TIE), Reviewer 2026
2026

Springer International Journal of Control, Automation, and Systems (IJCAS), Reviewer 2026
2026

International Conference on Control, Automation, and Systems (ICCAS), Reviewer 2026
2026

Projects

Development of an AI Robot Platform and Standard System for Seawater Pipeline Inspection Apr 2022 – Dec 2022
Developed robot platform technologies and standard systems for seawater pipeline inspection.
- Korea Hydro Nuclear Power
- Embedded Systems Engineering

Development of high-speed precision driving algorithm for intelligent active protection technology June 2023 – Dec 2023
Development of active protection technology system.
- Hanwha Aerospace
- Project Manager

DRB-KAIST SketchTheFuture Research Center July 2025 – present
Robotic immersive interaction platform for future mobility space design.
- DRB-Industrial
- Project Manager since 2025

Single-Person Multi-Purpose Autonomous Micro-Mobility for Factory and Campus Feb 2024 – present
Developed autonomous micro-mobility platforms and embedded control systems for factory and campus environments.
- DRB-Industrial
- Project Manager

Skills

Tools

Interests

Robotics

Languages

Korean
Native speaker

English
Intermediate